Declaration of contributions made by Michael John Hutchinson to joint work included in the dissertation "Generative modelling with diffusion models on Riemannian manifolds" for the degree of Doctor of Philosophy

The contents of this thesis is made up of four published papers, each turned into chapters.

These are:

- Chapter 3 is based on the paper "Riemannian Score-Based Generative Modelling".
- Chapter 4 is based on the paper "Diffusion Models for Constrained Domains".
- Chapter 5 is based on the paper "Metropolis Sampling for Constrained Diffusion Models".
- Chapter 6 is based on the paper "Geometric Neural Diffusion Processes".

The next pages declare my personal contributions to these works. In addition to these contributions I have rewritten large parts of the papers into chapter form.

1 Riemannian score-based generative models

1.1. Paper details.

- **Title of paper**: Riemannian Score-Based Generative Modelling.
- **Publication status**: Published.
- **Publication detail**:

  V. De Bortoli*, E. Mathieu*, M. J. Hutchinson*, J. Thornton, Y. W. Teh, and A. Doucet. Riemannian score-based generative modelling. *Advances in Neural Information Processing Systems*, 35, 2022.

  * indicates equal first author contribution

1.2. Student conformation.

- **Student name**: Michael John Hutchinson.
- **Contributions to the paper**:
  1. Project conception with Emile.
  2. Development of practical approach with Emile, Valentin.
  3. Development of the code base with Emile.
  4. Running of experiments: Earth and climate science, synthetic data on tori, and synthetic data on hyperbolic space experiments.
  5. Mathematical results: Implicit score matching on Riemannian manifolds.
  6. Original manuscript writing with Emile and Valentin.
- **Signature**:
- **Date**: 30/09/2024

1.3. Supervisor conformation.

- **Supervisor name and title**: Professor Yee Whye Teh.
- **Supervisor comments**:

  I confirm that Michael has contributed substantially to this project, as detailed above.

- **Signature**:
- **Date**: oct 2 2024

2 Diffusion Models for Constrained Domains

2.1. Paper details.

- **Title of paper**: Diffusion Models for Constrained Domains.
- **Publication status**: Published.
- **Publication detail**:

N. Fishman, L. Klarner, V. De Bortoli, E. Mathieu, and M. J. Hutchinson. Diffusion Models for Constrained Domains. *Transactions on Machine Learning Research*, 2023.

2.2. Student conformation.

- **Student name**: Michael John Hutchinson.
- **Contributions to the paper**:
  1. Project conception with Valentin, Nic.
  2. Development of practical approachs with Nic, Leo, Valentin, including the problem-specific details.
  3. Development of the code with Nic and Leo.
  4. Running experiments: Robotics experiment.
  5. Supervision of the project.
- **Signature**:
- **Date**: 30/09/2024

2.3. Supervisor conformation.

- **Supervisor name and title**: Professor Yee Whye Teh.
- **Supervisor comments**:

I confirm that Michael has contributed substantially to this project, as detailed above.

- **Signature**:
- **Date**: OCt 2 2024

3 Metropolis sampling for constrained diffusion models

3.1. Paper details.

- **Title of paper**: Metropolis sampling for constrained diffusion models.
- **Publication status**: Published.
- **Publication detail**:

  N. Fishman, L. Klarner, E. Mathieu, M. J. Hutchinson, and V. De Bortoli. Metropolis sampling for constrained diffusion models. *Advances in Neural Information Processing Systems*, 36, 2024.

3.2. Student conformation.

- **Student name**: Michael John Hutchinson.
- **Contributions to the paper**:
  1. Project conception with Valentin, Nic, Leo.
  2. Development of practical approachs with Nic, Leo, Valentin, including the problem-specific details.
  3. Development of the code with Nic and Leo.
  4. Running experiments: Robotics and convergence speed experiments.
  5. Supervision of the project.
- **Signature**:
- **Date**: 30/09/2024

3.3. Supervisor conformation.

- **Supervisor name and title**: Professor Yee Whye Teh.
- **Supervisor comments**:

  I confirm that Michael has contributed substantially to this project, as detailed above.
- **Signature**:
- **Date**: Oct 2, 2024

4 Geometric Neural Diffusion Processes

4.1. Paper details.

- **Title of paper**: Geometric Neural Diffusion Processes
- **Publication status**: Published.
- **Publication detail**:

  E. Mathieu*, V. Dutordoir*, M. J. Hutchinson*, V. De Bortoli, Y. W. Teh, and R. Turner. Geometric neural diffusion processes. *Advances in Neural Information Processing Systems*, 36, 2024.

  * indicates equal first author contribution

4.2. Student conformation.

- **Student name**: Michael John Hutchinson.
- **Contributions to the paper**:
  1. Project conception with Emile.
  2. Development of geometric nerual diffusion process framework with Emile, Vincent.
  3. Development of code with Emile, Vincent.
  4. Running of experiments: Global tropical cyclone trajectory prediction experiment.
  5. Developed the conditional sampling approaches.
  6. Mathematical results: Those regarding the invariance/equivariance of the proposed models, and the results regarding the equivariance of the posterior maps of invariant measures.
  7. Original manuscript writing with Emile and Vincent.
- **Signature**:
- **Date**: 30/09/2024

4.3. Supervisor conformation.

- **Supervisor name and title**: Professor Yee Whye Teh.
- **Supervisor comments**:

  I confirm that Michael has contributed substantially to this project, as detailed above.

- **Signature**:
- **Date**: Oct 2, 2024